

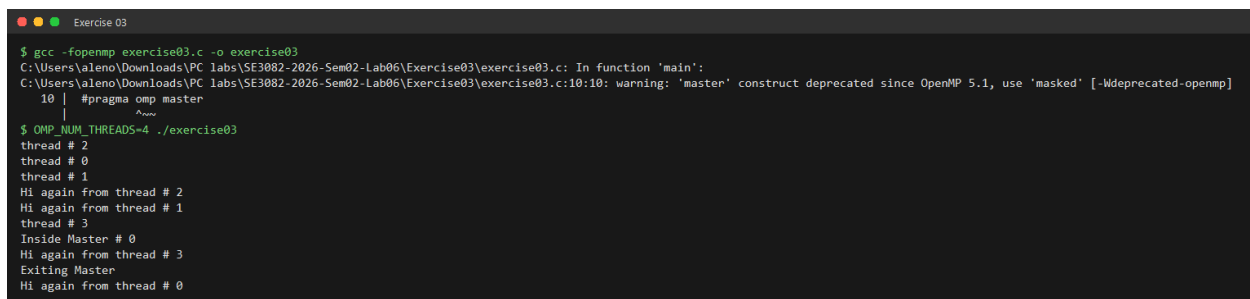
EXERCISE 3 - #pragma omp master

Files : exercise03.c

Compile / run

```
gcc -fopenmp exercise03.c -o exercise03
```

```
OMP_NUM_THREADS=4 ./exercise03
```



```
Exercise 03
$ gcc -fopenmp exercise03.c -o exercise03
C:\Users\aleno\Downloads\PC labs\SE3082-2026-Sem02-Lab06\Exercise03\exercise03.c: In function 'main':
C:\Users\aleno\Downloads\PC labs\SE3082-2026-Sem02-Lab06\Exercise03\exercise03.c:10:10: warning: 'master' construct deprecated since OpenMP 5.1, use 'masked' [-Wdeprecated-openmp]
   10 | #pragma omp master
      | ~~~~~
$ OMP_NUM_THREADS=4 ./exercise03
thread # 2
thread # 0
thread # 1
Hi again from thread # 2
Hi again from thread # 1
thread # 3
Inside Master # 0
Hi again from thread # 3
Exiting Master
Hi again from thread # 0
```

How omp master works in this program

- * The master block is executed ONLY by the master thread (thread 0), so "Inside Master # 0" is always printed by thread 0 and only once.
- * Unlike single, master has NO implicit barrier - the other threads do not wait. In the output thread 3 prints "Hi again" before the master block has even started, and the master lines are interleaved with the other threads' output.
- * gcc warns that master is deprecated since OpenMP 5.1 in favour of #pragma omp masked, which behaves the same way.